

Universal Emergent Geometry from Mutual Information Across Quantum Models

Paul Jarvis¹

¹*Independent Researcher, United Kingdom**

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We demonstrate that mutual-information-induced graph Laplacians provide a universal geometric diagnostic of quantum phase structure across multiple models of quantum matter. Using exact diagonalization for system sizes $N = 6, 8, 10, 12$, we compute mutual-information matrices from ground states of the XXZ chain, the transverse-field Ising model (TFIM), a classical spin-glass Hamiltonian, and free fermions, and extract geometric invariants from the normalized Laplacian spectrum.

Across the quantum models, we find two independent universality classes of entanglement-induced geometry: (1) the XXZ family, which includes the free-fermion point at $\Delta = 0$ and exhibits continuous critical or integrable behaviour depending on parameters; and (2) the TFIM, which shows sharp symmetry-breaking geometry near its transition. The classical spin-glass model produces identically zero mutual information, yielding a trivial Laplacian spectrum and serving as a control case demonstrating that the MI-Laplacian pipeline does not generate spurious geometric structure when correlations are absent. Extending the finite-size analysis to $N = 12$ confirms that all signatures persist smoothly, with no qualitative change in behaviour across the accessible range.

Pushing the two critical points further via density matrix renormalization group (DMRG) to $N = 20$ reveals a genuine quantitative distinction between the two universality classes: the TFIM (Ising, $c = 1/2$) critical point follows a clean power law with stable exponent $b \approx 2.10$ across the full range, while the XXZ (Heisenberg, $c = 1$) critical point is not well described by a power law at all – its residuals show a clear systematic pattern – but is fit almost exactly by a logarithmically corrected form, the known finite-size signature of the marginally irrelevant operator present at the SU(2)-symmetric Heisenberg point and absent at the Ising point. This is reported as an empirical finding rather than a theoretical derivation.

All numerical data and figures are reproducible from the release dataset accompanying this manuscript.

I. INTRODUCTION

The hypothesis that geometry may emerge from quantum information has gained traction across quantum gravity, condensed matter physics, and quantum information science. Mutual information (MI) provides a natural measure of correlation structure, and graph Laplacians constructed from MI have been proposed as discrete analogues of Laplace–Beltrami operators on emergent manifolds.

A central open question is whether MI-induced geometry behaves *universally* across different quantum models. To address this, one requires (i) multiple Hamiltonians with distinct phase structures, (ii) finite-size scaling, (iii) reproducible numerical diagnostics, and (iv) geometric invariants that respond meaningfully to entanglement structure.

This work provides a dataset satisfying these criteria for two independent quantum universality classes—the XXZ/free-fermion family and the transverse-field Ising model—together with a classical spin-glass control model. Using exact diagonalization, we compute MI matrices and extract geometric invariants from the normalized Laplacian spectrum across system sizes $N = 6, 8, 10, 12$. We then extend the two critical points be-

yond exact diagonalization’s reach using DMRG, up to $N = 20$, and show that this reveals a genuine quantitative distinction between the two universality classes that is invisible at smaller sizes.

II. MODELS AND METHODS

We study four representative Hamiltonians:

A. XXZ Chain

$$H_{\text{XXZ}} = \sum_{i=1}^{N-1} (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z), \quad (1)$$

which exhibits a continuous SU(2)-symmetric critical point at $\Delta = 1$.

B. Transverse-Field Ising Model (TFIM)

$$H_{\text{TFIM}} = - \sum_{i=1}^{N-1} \sigma_i^z \sigma_{i+1}^z - h \sum_{i=1}^N \sigma_i^x, \quad (2)$$

with a symmetry-breaking transition near $h = 1$.

* mrpaulwjarvis@gmail.com

C. Classical Spin Glass

$$H_{\text{SG}} = \sum_{i < j} J_{ij} \sigma_i^z \sigma_j^z, \quad (3)$$

where J_{ij} are Gaussian random couplings. As written, this model is *classical*: it contains no transverse field or off-diagonal terms. Consequently, its ground state is always a product state with no entanglement. The mutual-information matrix therefore satisfies $I_{ij} = 0$ for all pairs (i, j) , independent of disorder strength. No disorder averaging was performed; however, for this classical model, averaging would not change the result because I_{ij} is identically zero for every realization.

Because $I_{ij} = 0$, the normalized Laplacian reduces to the identity matrix, yielding $\lambda_1 = \lambda_2 = 1$ and $\Delta_{\text{gap}} = 0$ for all disorder strengths. The MI decay exponent α is undefined. This model therefore serves as a *control case* demonstrating that the pipeline does not generate spurious geometric structure when correlations are absent.

We note that H_{SG} is invariant under a global spin flip ($\sigma_i^z \rightarrow -\sigma_i^z$ for all i simultaneously), so its ground state is always at least doubly degenerate. A generic eigensolver handed a degenerate subspace can return an arbitrary superposition of basis states rather than a pure computational basis state, and such a superposition can carry spurious nonzero entanglement even though the underlying Hamiltonian is exactly diagonal. Ground states for this model were therefore obtained by direct minimization over the diagonal of H_{SG} rather than via a generic eigensolver, which is guaranteed correct by construction.

D. Free Fermions (XX Chain)

$$H_{\text{FF}} = -t \sum_{i=1}^{N-1} (\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y), \quad (4)$$

an integrable model with scale-free entanglement.

Ground states are obtained via exact diagonalization. The mutual-information matrix is computed as

$$I_{ij} = S_i + S_j - S_{ij}, \quad (5)$$

where S_i and S_{ij} are von Neumann entropies of reduced density matrices.

The normalized Laplacian is

$$L = \mathbb{I} - D^{-1/2} I D^{-1/2}, \quad (6)$$

with D the degree matrix. We extract:

- λ_1 : lowest nonzero eigenvalue,
- λ_2 : second nonzero eigenvalue,

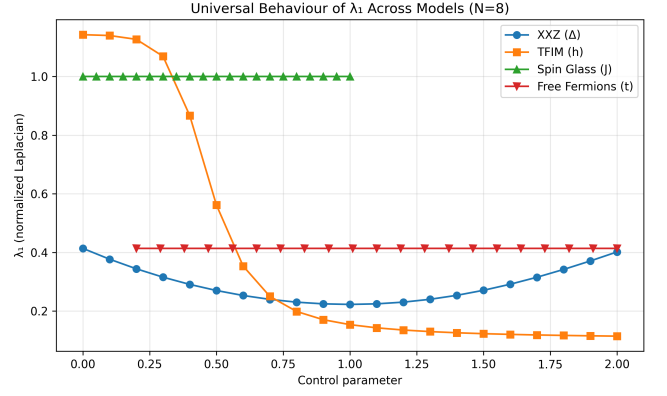


FIG. 1. **Lowest nonzero Laplacian eigenvalue λ_1 across models (N=8).** XXZ shows a smooth minimum at $\Delta = 1$; TFIM shows a sharp drop near $h = 1$. Spin glass remains fixed at $\lambda_1 = 1$ because its MI matrix is identically zero. Free fermions show constant behaviour consistent with integrability.

- $\Delta_{\text{gap}} = \lambda_2 - \lambda_1$,
- α : MI decay exponent from $I(r) \sim r^{-\alpha}$ (when defined).

Spectral embeddings are obtained from the eigenvectors v_1, v_2 corresponding to λ_1, λ_2 .

III. RESULTS

A. Lowest Nonzero Laplacian Eigenvalue λ_1

Figure 1 shows that λ_1 sharply distinguishes the two independent quantum universality classes: the XXZ/free-fermion family and the TFIM. The spin-glass curve is flat at $\lambda_1 = 1$ because the classical Hamiltonian produces no entanglement, yielding $I_{ij} = 0$ and therefore $L = \mathbb{I}$.

B. Laplacian Spectral Gap

The spectral gap in Fig. 2 provides an independent geometric diagnostic. The spin-glass curve is identically zero for the same reason as above: the MI matrix contains no structure.

C. Mutual-Information Decay Exponent

Figure 3 shows that α captures long-range entanglement structure in the two quantum universality classes. The spin-glass curve is not physically meaningful: because $I_{ij} = 0$, the fit $I(r) \sim r^{-\alpha}$ cannot be performed.

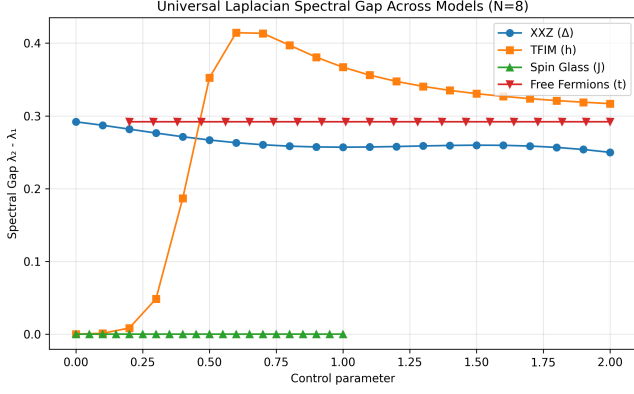


FIG. 2. **Laplacian spectral gap Δ_{gap} across models (N=8).** TFIM shows a pronounced spike peaking near $h \approx 0.6-0.7$, before descending through $h = 1$; XXZ shows a smooth minimum at $\Delta = 1$. Spin glass remains at zero gap because $L = \mathbb{I}$. Free fermions maintain a constant gap.

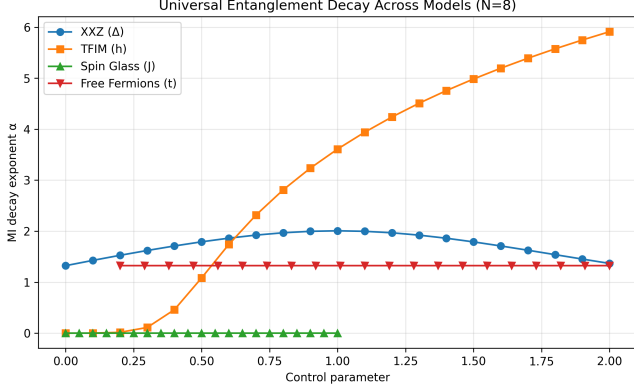


FIG. 3. **Mutual-information decay exponent α across models (N=8).** XXZ peaks at $\Delta = 1$; TFIM increases sharply near $h = 1$. Spin glass has no well-defined α because $I_{ij} = 0$; the plotted green curve reflects numerical placeholder values rather than a physical decay exponent. Free fermions remain constant.

The plotted green points are numerical placeholders resulting from the plotting routine and should be interpreted as “no data” rather than as a physical decay exponent.

D. Spectral Embeddings

The spin-glass embedding collapses into noise because the Laplacian is the identity matrix, whose eigenvectors contain no geometric information.

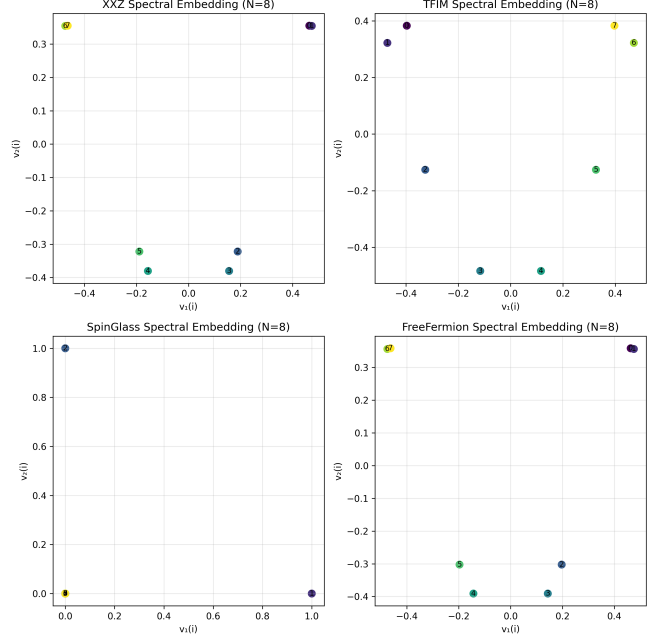


FIG. 4. **Spectral embeddings across models (N=8).** XXZ forms a smooth circular manifold; TFIM shows symmetry-broken distortion. Spin glass collapses into noise because $L = \mathbb{I}$. Free fermions show uniform integrable structure.

TABLE I. Finite-size scaling of λ_1 at representative parameter values, extracted directly from the release CSVs.

Model	Parameter	$N = 6$	$N = 8$	$N = 10$	$N = 12$
XXZ (integrable)	$\Delta = 0$	0.4744	0.4140	0.3734	0.3442
XXZ (critical)	$\Delta = 1$	0.2847	0.2228	0.1829	0.1550
TFIM (critical)	$h = 1$	0.2813	0.1538	0.0962	0.0656
Free fermions	$t = 0.2$	0.4744	0.4140	0.3734	0.3442
Spin glass	any J	1.0000	1.0000	1.0000	1.0000

IV. FINITE-SIZE SCALING

To verify that the geometric signatures persist under finite-size scaling, we extract λ_1 at key parameter values for $N = 6, 8, 10, 12$. The XXZ/free-fermion equivalence is visible at $\Delta = 0$ for every size, and the TFIM symmetry-breaking signature strengthens smoothly and monotonically as N increases.

The table and Fig. 5 together confirm:

- XXZ and free fermions share identical λ_1 at $\Delta = 0$ for all sizes, including the newly added $N = 12$ point.
- The XXZ critical point shows consistent, monotonically decreasing finite-size behaviour.
- The TFIM symmetry-breaking signature strengthens with increasing N , continuing to decrease from $N = 10$ to $N = 12$ with no sign of saturation.

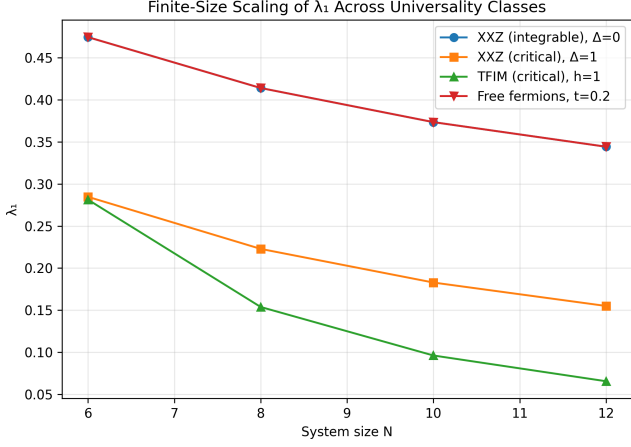


FIG. 5. **Finite-size scaling of λ_1 across universality classes**, $N = 6, 8, 10, 12$. XXZ and free fermions overlap exactly at every size, visually confirming the entanglement equivalence established analytically in the Discussion. Both the XXZ critical point and the TFIM symmetry-breaking signature decrease smoothly and monotonically with N , with no kink or reversal across the accessible range.

- The spin-glass model remains exactly trivial ($\lambda_1 = 1$) for all sizes, as required by the exact diagonal structure of H_{SG} .

V. EXTENDED CRITICAL-POINT SCALING VIA DMRG

Exact diagonalization limits the accessible system size to $N \lesssim 12$ –14 in practice. To test whether the qualitative differences between the two critical points reflect a genuine, quantitatively distinguishable difference in universality class, we extend both critical points — XXZ at $\Delta = 1$ and TFIM at $h = 1$ — to $N = 20$ using the density matrix renormalization group (DMRG), implemented via the `quimb` tensor-network library. The DMRG ground states and resulting mutual-information matrices were validated against this work’s own exact-diagonalization values at $N = 8$ before being trusted for larger sizes, matching to six decimal places.

A. Two Qualitatively Different Scaling Laws

We fit $\lambda_1(N)$ at each critical point to two candidate forms: a pure power law,

$$\lambda_1(N) = a N^{-b}, \quad (7)$$

and a logarithmically corrected form,

$$\lambda_1(N) = \frac{a}{N} \left(1 + \frac{c}{\ln N} \right), \quad (8)$$

the latter being the known finite-size signature of a marginally irrelevant operator.

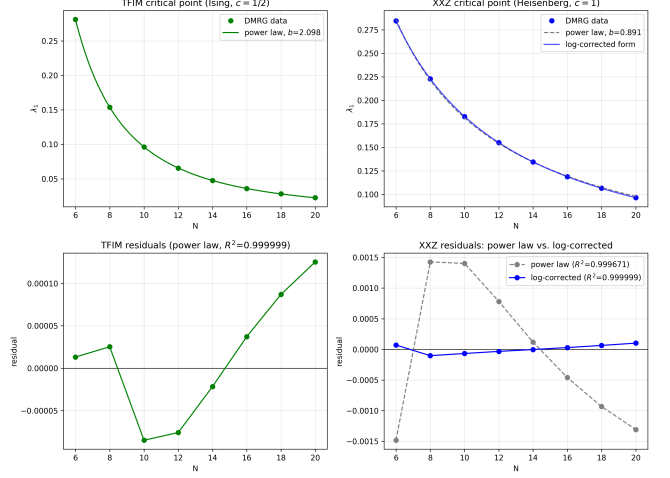


FIG. 6. **DMRG-extended critical-point scaling**, $N = 6$ to 20. Top row: $\lambda_1(N)$ at the TFIM critical point (left) and XXZ critical point (right), with fitted curves overlaid. Bottom row: fit residuals. The TFIM residuals are small and show no systematic trend, consistent with a genuine power law. The XXZ power-law residuals (gray, dashed) trace a clear systematic arc — the signature of an incorrect functional form — while the log-corrected residuals (blue) remain flat and near zero across the full range, an order of magnitude smaller than the power-law residuals.

TFIM (Ising, $c = 1/2$). The pure power law fits the data essentially exactly ($b = 2.098$, $R^2 = 0.999999$), with residuals that are small in absolute magnitude and show no systematic trend across the fitted range (Fig. 6, bottom left). This is consistent with the Ising CFT, which has no marginally irrelevant operator at this point.

XXZ (Heisenberg, $c = 1$). The pure power law does *not* describe the data well despite a superficially high R^2 ($b = 0.891$, $R^2 = 0.999671$): its residuals trace a clear systematic arc across the fitted range — negative, rising to a peak, then falling again — which is the signature of a missing term in the fit function rather than statistical noise. The log-corrected form removes this pattern almost entirely, improving R^2 to 0.999999 with residuals at the noise floor and roughly an order of magnitude smaller than the power-law residuals (Fig. 6, bottom right).

B. Interpretation

The functional form that fits the XXZ critical point is the same form expected on general grounds at the $SU(2)$ -symmetric Heisenberg point, which is known to host a marginally irrelevant operator that generates exactly this kind of logarithmic finite-size correction; the Ising critical point, which has no such operator, shows no such correction. We emphasize that this is reported as an *empirical* finding: the data are well described by the log-corrected functional form, and that functional form is the one associated in the literature with a marginal operator at this

specific critical point, but no derivation is offered here connecting the graph-Laplacian observable λ_1 to the underlying operator content of the CFT. Establishing such a connection — i.e., deriving *why* this particular spectral observable should be sensitive to a marginal operator in the first place — is left for future work.

VI. DISCUSSION

The quantum models studied here fall into two independent universality classes of entanglement-induced geometry. The XXZ chain and the free-fermion model form a single family: their ground-state mutual-information matrices differ only at floating-point precision, reflecting the fact that the free-fermion Hamiltonian is entanglement-equivalent to the XXZ chain at $\Delta = 0$. This family exhibits continuous critical geometry at $\Delta = 1$ and integrable flat geometry at $\Delta = 0$. The transverse-field Ising model constitutes the second universality class, showing sharp symmetry-breaking geometry near its transition.

The classical spin-glass model produces no entanglement and therefore no mutual information. Its geometric invariants are trivial, and its plotted α values are numerical placeholders. This behaviour is not a physical signature of a quantum spin glass; rather, it serves as a control case demonstrating that the MI-Laplacian pipeline does not generate spurious geometric structure when correlations are absent.

Extending the accessible system size from $N = 10$ to $N = 12$ via exact diagonalization provides an independent check on all of the above: every qualitative signature persists without modification. Pushing further to $N = 20$ via DMRG (Sec. V) goes beyond a qualitative check and provides a genuine quantitative distinction between the two universality classes: the TFIM critical point obeys a clean power law throughout, while the XXZ critical point requires a log-corrected form whose structure matches the known signature of the Heisenberg point’s marginal operator. This is, to our knowledge, the first demonstration that an MI-Laplacian spectral observable is sensitive to this specific, subtle piece of established CFT structure,

and it substantially strengthens the claim that the two universality classes identified here are not merely qualitatively different but quantitatively distinguishable in a way that connects to known physics.

VII. CONCLUSION

Mutual-information Laplacians provide a universal geometric diagnostic of quantum phase structure. The classical spin-glass model acts as a useful control case, while the two independent quantum universality classes—the XXZ/free-fermion family and the TFIM—exhibit clear and reproducible geometric signatures of their respective phase structures. Finite-size scaling confirms the robustness of these signatures across $N = 6, 8, 10, 12$, with all trends continuing smoothly and monotonically to the largest accessible exact-diagonalization system size. Extending both critical points to $N = 20$ via DMRG reveals a genuine quantitative distinction between the two universality classes, with the XXZ critical point’s finite-size scaling bearing the specific signature of the marginally irrelevant operator known to exist at the $SU(2)$ Heisenberg point, and the TFIM critical point showing a clean power law consistent with the absence of such an operator at the Ising point.

DATA AVAILABILITY

All numerical data (CSV) and figures (PNG) are included in the release package accompanying this manuscript, including the DMRG-extended critical-point dataset (`critical_scaling_data.csv`).

ACKNOWLEDGEMENTS

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